



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PART TEST

31-12-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE-S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	A	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	A,B,C,D	A,B	C	A,B	B,C	A	A,B
SECTION-II	Q.	1	2	3	4	5	6		
	A.	95.00	4.80	16.00	21.40 to 21.50	0.14 to 0.15	3.00		

PART-2 : CHEMISTRY

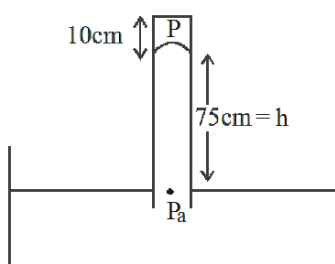
SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	A	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C	B,C,D	A,C,D	A,B,C,D	A,B,C,D	B,C,D	C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.00	44.40	1.00	3.00	6.00	3.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	A	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B	A,C	A,B,C,D	B,C,D	A,B,C	A,D	A,C,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	6.00	0.00	10.00	2.50	1.10	3.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A)



Let the pressure at the top of mercury be P

$$P = P_a - h\rho g - \frac{2T \cos \theta}{r}$$

$$P = 76 \text{ cm of Hg} - 75 \text{ cm of Hg} - \frac{2T \cos \theta}{r}$$

$$= 1 \times 10^{-2} \times 13.6 \times 10^3 \times 10 - \frac{2 \times 0.5}{\sqrt{2} \times 10^{-3}} \times \frac{1}{\sqrt{2}}$$

$$= 860 \text{ N/m}^2$$

No of moles of gas

$$= n = \frac{PV}{RT} = \frac{860 \times \pi (\sqrt{2} \times 10^{-3})^2}{R \times 310}$$

$$= \frac{0.55\pi}{R} \times 10^{-6}$$

2. Ans (C)

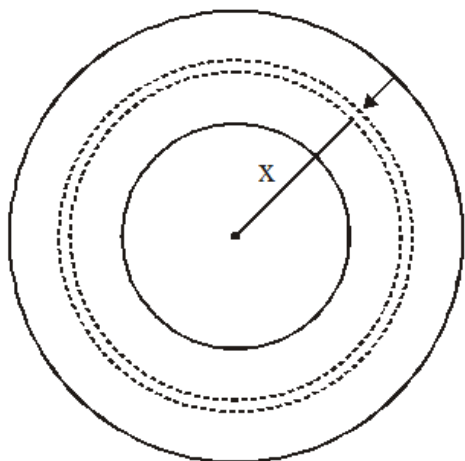
Due to weight there is tensile force and due to thermal expansion there is compressive stress.

 Maximum tensile stress is at top = $\frac{W_0}{A}$

 Compressive stress due to expansion is = $Y\alpha\Delta T$

$$\text{So, } Y\alpha\Delta T \geq \frac{W_0}{A}, \Delta T \geq \frac{W_0}{Y\alpha A}$$

3. Ans (A)



Consider after time t a spherical region of radius r is of water at the core.

Thermal resistance of the spherical ice piece for radial flow is

$$R(r) = \int dR = \int_r^a \frac{dr}{k4\pi r^2}$$

$$R(r) = \frac{1}{4\pi k} \left[\frac{1}{r} - \frac{1}{a} \right]$$

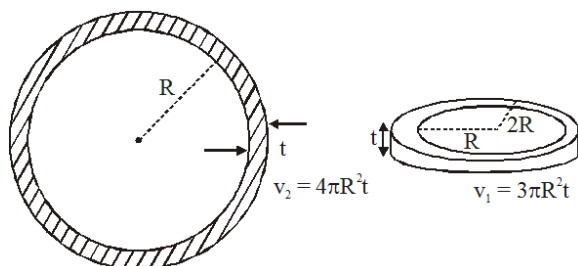
Radial rate of flow of heat H

$$H = \frac{\theta}{R(r)} = -\rho L_f 4\pi r^2 \frac{dr}{dt}$$

$$\Rightarrow \int_0^t \frac{\theta}{\rho L_f} dt = \int_a^{a/2} 4\pi r^2 \frac{d}{4\pi k} \left[\frac{1}{r} - \frac{1}{a} \right]$$

$$t = \frac{\rho L_f}{k\theta} \frac{a^2}{12}$$

4. Ans (D)



Comparing rate of cooling of

$$\frac{dT}{dt} = \frac{eA\sigma(T^4 - T_0^4)}{ms} \propto \frac{A}{ms} \propto \frac{A}{\rho V S}$$

$$\frac{4\pi R^2}{\rho_A 4\pi R^2 t S_A} = \frac{6\pi R^2}{\rho_B 3\pi R^2 t S_B}$$

$$2\rho_A S_A = \rho_B S_B$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$$\phi_B = hf_0 \quad hf_0 - chf_0 = 2.4\text{eV}$$

$$\phi_A = hcf_0 \quad \phi_B - \phi_A = 2.4\text{eV}$$

$$(hf - chf_0) - (hf - hf_0) = 2.4\text{eV}$$

$$hf_0 = \frac{1242}{300} = 4.14\text{eV}$$

$$\text{Also } hf_0(1 - c) = 2.4\text{ eV}$$

$$\text{So } C = 0.42 \text{ and } \phi_A = Chf_0 = 1.74\text{ eV}$$

$$\text{In second experiment } 1.2 hf_0 - \phi = eV_s$$

$$\text{For sample A} \rightarrow V_s = \frac{1.2hf_0 - \phi_A}{e} = 3.18\text{ volts}$$

$$\text{For sample B} \rightarrow V_s = \frac{1.2hf_0 - \phi_B}{e} = 0.83\text{ volts}$$

6. Ans (A,B,C,D)

$$\text{Bohr's orbit radius } r_n = \frac{n^2 R_0}{Z} = \frac{n^2}{Z} 0.53\text{\AA}$$

$$r_{n_1} = 47.7\text{\AA}, r_{n_2} = 84.8\text{\AA}, Z = 10$$

$$\text{So, } n_1 = 30, n_2 = 40$$

$$\frac{F_1}{F_2} = \left(\frac{r_2}{r_1} \right)^2 = \left(\frac{n_2}{n_1} \right)^4$$

$$\frac{k_1}{k_2} = \left(\frac{n_2}{n_1} \right)^2, E_n = -13.6 \frac{Z^2}{n^2}$$

$$\Delta E = 13.6 \times 100 \left[\frac{1}{900} - \frac{1}{1600} \right] = 0.66\text{eV}$$

$$|k_1 - k_2| = |E_1 - E_2|$$

7. Ans (A,B)

$$R_0(1 - e^{-kt}) - \lambda N = \frac{dN}{dt}$$

$$\frac{dN}{dt} + \lambda N = R_0(1 - e^{-kt})$$

$$e^{\lambda t} \frac{dN}{dt} + \lambda N e^{\lambda t} = R_0(1 - e^{-kt}) e^{\lambda t}$$

$$\int d(Ne^{\lambda t}) = \int R_0(e^{\lambda t} - e^{(\lambda-k)t}) dt + C$$

$$Ne^{\lambda t} = R_0 \frac{e^{\lambda t}}{\lambda} - \frac{R_0 e^{(\lambda-k)t}}{\lambda - k} + C$$

$$t = 0, N = 0$$

$$\frac{R_0}{\lambda} - \frac{R_0}{\lambda - k} + C = 0$$

$$C = \frac{R_0}{\lambda - k} - \frac{R_0}{\lambda}$$

$$Ne^{\lambda t} = R_0 \frac{e^{\lambda t}}{\lambda} - \frac{R_0 e^{(\lambda-k)t}}{\lambda - k} + \frac{R_0}{\lambda - k} - \frac{R_0}{\lambda}$$

$$N = \frac{R_0}{\lambda} - \frac{R_0}{\lambda} e^{-\lambda t} + \frac{R_0}{\lambda - k} e^{-\lambda t} - \frac{R_0}{\lambda - k} e^{-kt}$$

$$= \frac{R_0}{\lambda} (1 - e^{-\lambda t}) + \frac{R_0}{\lambda - k} [e^{-\lambda t} - e^{-kt}]$$

8. **Ans (C)**

Work done by the gas is negative in CA, zero in BC and positive in AB.

Temperature of gas first increases then decreases during AB.

Temperature of gas decreases during CA as PV decreases.

$$\begin{aligned} Q_{CA} &= W_{CA} + \Delta U_{CA} \\ &= -1.25 \times 3 \times 10^5 - \frac{5}{2} n R \Delta t \\ &= -3.75 \times 10^5 - \frac{5}{2} (P_2 V_2 - P_1 V_1) \\ &= -3.75 \times 10^5 - \frac{5}{2} \times 5 \times 10^5 \end{aligned}$$

9. **Ans (A, B)**

Let m_1 be the amount of ice and m_2 be the mass of water.

$$m_1 \times 335 + m_1 \times 4.2 \times 45 = P \times t = 10 \text{ (in kJ)}$$

$$m_1 = \frac{10}{524} \text{ kg} = 19 \text{ gm}$$

$$m_2 \times 4.2 \times 25 = 50 \text{ (in kJ)}$$

$$m_2 = \frac{1}{2.1} \text{ kg} = 476 \text{ gm}$$

$$m_1 + m_2 = 0.495 \text{ kg}$$

$$0.495 \times 4.2 \times \Delta \theta = 20$$

$$\Delta \theta = 9.6^\circ \text{C}$$

$$\text{Find temperature } 45 + 9.6 = 54.6^\circ \text{C}.$$

10. **Ans (B, C)**

Ratio of area of cross-section

$$A_A : A_B : A_C = 1 : \frac{3}{2} : 2$$

$$\text{stress} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Ratio} \Rightarrow \frac{mg}{A_A} : \frac{\frac{3}{2}mg}{A_B} : \frac{\frac{mg}{2}}{A_C} \Rightarrow \frac{1}{1} : \frac{\frac{3}{2}}{\frac{3}{2}} : \frac{\frac{1}{2}}{2}$$

$$4 : 4 : 1$$

$$\text{compression} = \frac{F \ell}{A y}$$

$$\text{Ratio} \Rightarrow 4 \times 1 : 4 \times \frac{3}{2} : 1 \times \frac{1}{2}$$

$$8 : 12 : 1$$

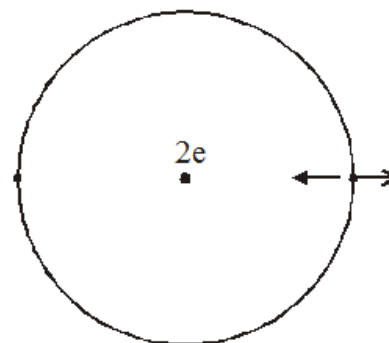
$$\frac{\text{energy}}{\text{volume}} = \frac{(\text{Stress})^2}{2y} \quad \text{Ratio} \Rightarrow 16 : 16 : 1$$

$$\text{total energy} = \frac{1}{2} \text{ force} \times \text{compression}$$

$$\text{Ratio} \Rightarrow 1 \times 8 : \frac{3}{2} \times 12 : \frac{1}{2} \times 1$$

$$16 : 36 : 1$$

11. **Ans (A)**



$$\frac{m^2 v^2 n^2}{m v^2} = \frac{\frac{n^2 h^2}{4 \Lambda^2}}{\frac{7}{4} \frac{ke^2}{r}}$$

$$m r^2 = \frac{n^2 h^2 r}{7 \pi^2 k e^2}$$

$$r = \frac{n^2 h^2}{7 m e^2 \pi^2 k} = \frac{4 n^2 h^2 \epsilon_0}{7 m e^2 \pi}$$

$$F = \frac{k z e^2}{r^2} = \frac{ke^2}{4 r^2}$$

$$\frac{ke^2}{r} \left[2 - \frac{1}{4} \right] = \frac{m v^2}{r}$$

$$\frac{7}{8} \frac{ke^2}{r} = \frac{1}{2} m v^2 = k E$$

$$U = \frac{-k 2 e^2}{r} + \frac{ke^2}{2r} = \frac{-ke^2}{r} \left[2 - \frac{1}{2} \right]$$

$$= -\frac{3}{2} \frac{ke^2}{r}$$

$$E = \frac{7}{8} \frac{ke^2}{r} - \frac{12}{8} \frac{ke^2}{r}$$

$$E = -\frac{5}{8} \frac{ke^2}{r} = -\frac{5}{8} \frac{1}{4 \pi \epsilon_0} \frac{e^2 7 \pi m e^2}{4 h^2 \epsilon_0} = \frac{-35 m e^4}{128 n^2 h^2 \epsilon_0^2}$$

12. **Ans (A, B)**

For adiabatic process $P^{1-\gamma} T^\gamma = \text{const.}$

$$r = \frac{C_P}{C_V} = \frac{C_V + R}{C_V} = \frac{615 + 410}{615} = \frac{5}{3}$$

$$P \propto T^{\frac{-\gamma}{1-\gamma}}$$

$$\Rightarrow P_2 = P_1 \left(\frac{363}{300} \right)^{\frac{5}{2}} = 322 \text{ kPa}$$

Now heat exchange between gas and block.

$$1 \times 615 \times (363 - x) = 2 \times 305 \times (x - 300)$$

$$615 \times 363 - 615 x = 610 x - 183000$$

$$1225 x = 406245$$

$$x = 331.6 \text{ K}$$

$P \propto T$ for constant volume.

$$P_{\text{final}} = 322 \text{ kPa} \times \left(\frac{332}{363} \right) = 294 \text{ kPa}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (95.00)**

$$K_0 (1 + \alpha T) A \frac{dT}{dx} = H$$

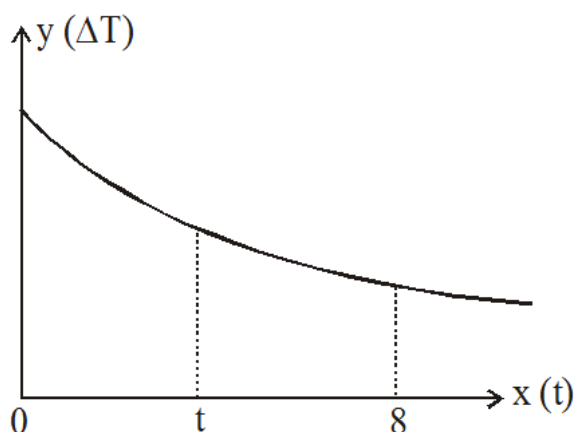
$$\int_{300}^{500} K_0 (1 + \alpha T) A dT = \int_0^{\ell} H dx = H\ell \dots (1)$$

$$\int_{410}^{500} K_0 (1 + \alpha T) A dT = \int_0^{\ell/2} H dx = H \frac{\ell}{2} \dots (2)$$

Dividing 1 & 2 and solving the integral we get

$$\alpha = \frac{1}{95}$$

2. **Ans (4.80)**



Taking $t = 0$ as 12 noon.

$$\frac{dT}{dt} = -k(T - 72) \rightarrow 2 = -k(80^\circ - 72^\circ) \rightarrow k = \frac{1}{4}$$

$$\Delta T = \Delta T_0 e^{-\frac{1}{4}t}$$

$$8 = \Delta T_0 e^{-\frac{9}{4}} \rightarrow \Delta T_0 = 8e^{\frac{9}{4}}$$

$$\Delta T = 8e^{\frac{9}{4}} \cdot e^{-\frac{t}{4}} = 8e^{\frac{9-t}{4}}$$

$$26.6 = 8e^{\left(\frac{9-t}{4}\right)} \rightarrow e^{\frac{9-t}{4}} = \frac{26.6}{8}$$

$$t = 9 - 4 \ln\left(\frac{26.6}{8}\right)$$

$$t = 9 - \ln(3.325)$$

$$= 9 - 4 \times 1.20$$

$$= 4.2 \text{ h}$$

$$= 4 \text{ h } 12 \text{ min past 12 noon.}$$

3. **Ans (16.00)**

let O_2 has n moles.

Molar specific heat of mixture initially = C_{v1}

$$C_{v1} = \frac{n \times \frac{5}{2}R + \frac{1}{2} \times \frac{5}{2}R}{n + \frac{1}{2}} = \frac{5}{2}R$$

$$C_{v2} = \frac{2n \times \frac{3}{2}R + \frac{1}{2} \times \frac{5}{2}R}{2n + \frac{1}{2}}$$

$$C_{v1} - C_{v2} = \frac{2}{3}R$$

Solving we get $n = \frac{1}{2}$, $M \rightarrow 16 \text{ gm.}$

4. **Ans (21.40 to 21.50)**

$$Q = BE(\text{products}) - BE(\text{reactants})$$

$$= 7.6 \text{ 0} + 234 + 7.1 \times 4 - 7.5 \times 238$$

$$= 1778.4 + 28.4 - 1785$$

$$= 21.8 \text{ MeV}$$

$$KE_\alpha = Q \frac{234}{238} = 21.43 \text{ MeV}$$

5. **Ans (0.14 to 0.15)**

$$2 \times 1.75 \times 3.5 \times 10 = 0.2 \times 4200 \times \Delta T$$

$$\Delta T = 0.146^\circ\text{C}$$

6. **Ans (3.00)**

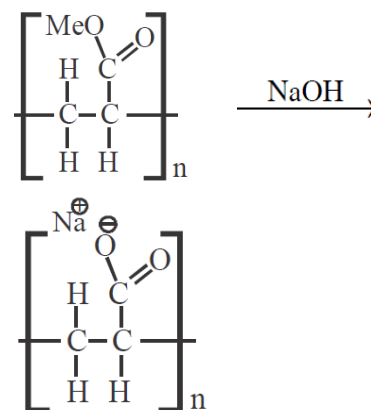
$$\text{Shear stress} = \frac{150 \times 10^3}{2\pi \times 10 \times 10^{-3} \times 8 \times 10^{-3}}$$

$$= 3 \times 10^8 \text{ N/m}^2$$

PART-2 : CHEMISTRY

SECTION-I (i)

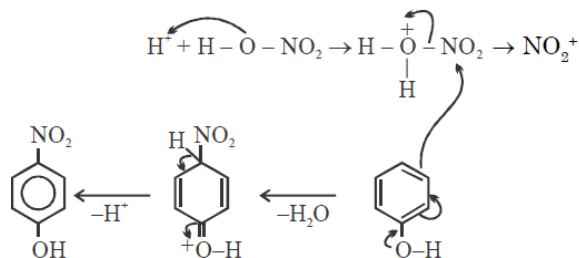
1. **Ans (C)**



Poly(methyl acrylate)

Sodium polyacrylate

2. Ans (D)



3. Ans (A)



PH_4I exists as $PH_4^+ I^-$

Hybridization of 'P' in PH_4^+ is sp^3

4. Ans (D)

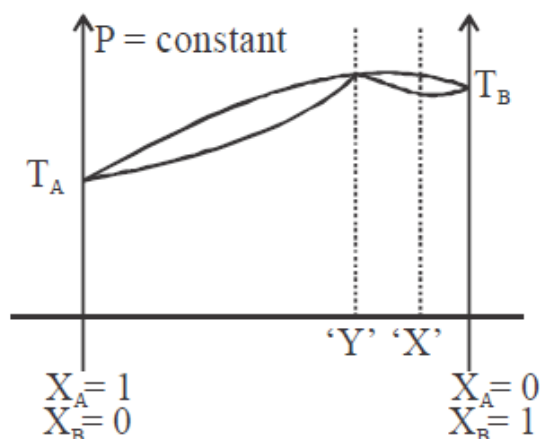
Option (a), (b), (c) are NCERT based.

(d) $\rightarrow$ Heterogeneous catalysis involves different phase of reactant and catalyst.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,C)



(d) If external pressure changes, azeotropic point also changes i.e., 'Y' changes.

(c) At 'X' composition, B.P. is greater than T_B .

(b) -ve deviation solution forms maximum B.P. azeotropic mixture.

6. Ans (B,C,D)

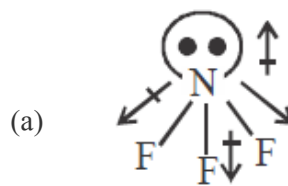
NCERT based

7. Ans (A,C,D)

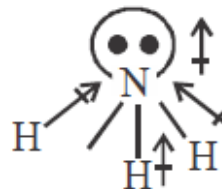
Bond order is zero for

He_2 , Be_2 and HeH^-

8. Ans (A,B,C,D)

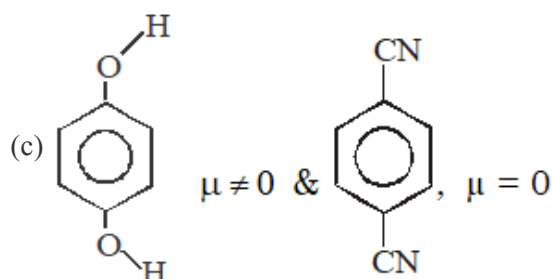


Net cancellation of dipoles

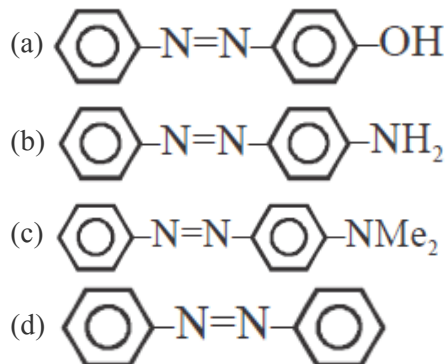


Net addition of dipoles

(b) $CHCl_3 < CH_2Cl_2$

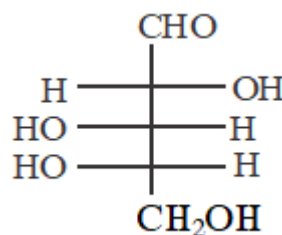


9. Ans (A,B,C,D)



10. Ans (B,C,D)

Correct structure of L-Arabinose is



11. **Ans (C,D)**

(a) $i < 1$, hence, F.P. $\downarrow$ is less

(b) $i < 1$, hence, F.P. $\downarrow$ is less

(c) $i > 1$, hence, F.P. $\downarrow$ is more than expected

(d) $i > 1$, hence, F.P. $\downarrow$ is more than expected

$$\Delta T_f = i k_f m$$

(b) Milk is a colloidal sol

12. **Ans (A,B,C)**

(a) and (b) are gas in solid $\rightarrow$ solid sol

(c) is solid in solid $\rightarrow$ solid sol

(d) is gas in liquid $\rightarrow$ foam

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (1.00)**

(a) Area occupied by each molecule

$$= 1 \times 4 \times 10^{-20} \text{ cm}^2$$

(b) Maximum molecules that can be accommodated per cm^2 of Pt-surface

$$= \frac{0.1 \times 10^{-6}}{60} \times 6 \times 10^{23} = 10^{15}$$

(a) $\Rightarrow$ Max no. of molecules of 'X' in 1 cm^2 area

$$= \frac{1}{4 \times 10^{20}} = 2.5 \times 10^{19} \text{ But, Pt - surface can}$$

only accommodate 10^{15} molecules in 1 cm^2

[From (b)]

Hence, total no. of molecules accommodated on

$$20 \times 30 \text{ cm}^2 = 10^{15} \times 20 \times 30 = 6 \times 10^{17}$$

$$\therefore \text{No. of moles} = \frac{6 \times 10^{17}}{6 \times 10^{23}} = 10^{-6}$$

No. of μ -moles = 1

2. **Ans (44.40)**

V.P. due to water in 95 mol % solution

$$= 0.95 \times 19$$

$$= 17.1 \text{ torr}$$

$$P(\text{H}_2\text{S}) = 40 - 17.1 = 22.9 \text{ torr}$$

$$\text{Henry's constant, } K_H(\text{H}_2\text{S}) = \frac{22.9}{0.05} = 458 \text{ torr}$$

$$\text{Hence, } P(\text{H}_2\text{S}) \text{ for } (6 \text{ mol \% solution}) = 458 \times 0.06 = 27.48$$

$$P(\text{H}_2\text{O}) = 0.94 \times 18 = 16.92$$

$$\therefore \text{Total V.P.} = 27.48 + 16.92 = 44.4 \text{ torr}$$

3. **Ans (1.00)**

Threonine has 2 chiral centre

4. **Ans (3.00)**

NO , KO_2 , ClO_3 are odd e^- species.

5. **Ans (6.00)**

H_2O_2 , K_3CrO_8 , HNO_4 , H_3PO_5 , $\text{H}_2\text{S}_2\text{O}_8$, H_2SO_5

Have peroxy linkage

6. **Ans (3.00)**

Only H_3BO_3 , Cl_2O and CO satisfy the above criteria.

$\text{H}-\text{O}-\text{H}$ & $\text{F}-\text{O}-\text{F}$ are planar but lack back bond

$-(\text{CH}_3)_2\text{O}$ is non planar and lacks backbond

$-(\text{SiH}_3)_2\text{O}$ has back bonding from 'O' but lacks planarity about Si.

$-\text{CH}_3-\text{N}=\text{C}=\text{O}$ is non-planar about sp^3 'C'.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \text{ for each } a_{ij} > 0$$

$$\text{adj } A = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}^T, \text{ where } c_{ij} \text{ denotes}$$

cofactor of a_{ij} . If we take any two c_{ij} from any row or column.

$$\text{Let } c_{11} = 0 \quad c_{12} = 0$$

$$a_{22} \cdot a_{33} - a_{32} \cdot a_{23} = 0$$

$$\frac{a_{22}}{a_{32}} = \frac{a_{23}}{a_{33}} \quad \dots(i)$$

$$\text{For } c_{12} = 0$$

$$a_{21} \cdot a_{33} - a_{31} \cdot a_{23} = 0$$

$$\frac{a_{21}}{a_{31}} = \frac{a_{23}}{a_{33}} \quad \dots(ii)$$

From (i) and (ii) we get

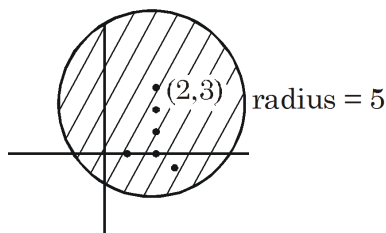
$$\frac{a_{21}}{a_{31}} = \frac{a_{22}}{a_{32}} = \frac{a_{23}}{a_{33}} \Rightarrow |A| = 0$$

It is not possible $\Rightarrow$ at most one $c_{ij} = 0$ from each row or column.

2. **Ans (A)**

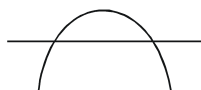
Let $z + 2 = z'$

$|z' - 2 - 3i| \leq 5$



$\Rightarrow |z'|_{\min} = 0$

3. **Ans (C)**



$D > 0$

$b^2 - 4a^2c > 0$

$b^2 > 4a^2c$

4. **Ans (A)**

Let a = sum of n terms of GP

b = product of n terms of GP

c = sum of reciprocals of n terms of GP

Given the result is $b^2 = \left(\frac{a}{c}\right)^n$

Here $n = 37 \Rightarrow b^2 = \left(\frac{a}{c}\right)^{37}$

$m = 2, n = 37$

So, the last digit of $2^{37} + 37^2$ is 1.

PART-3 : MATHEMATICS

SECTION-I (ii)

6. **Ans (A,C)**

$|Z_1| = 4, |Z_2| = 2$ and $2Z_2 = Z_1$

$\omega = e^{\frac{i2\pi}{21}}$ is 21th roots of unity.

$\omega + \omega^2 + \dots + \omega^{20} = -1$

Now, $\sum_{r=1}^{20} |z_1 + \omega^r z_2|^2$

$= \sum_{r=1}^{20} (|z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_1 \bar{z}_2 \bar{\omega}^r))$

$= 20|z_1|^2 + 20|z_2|^2 + 2 \sum_{r=1}^{20} \operatorname{Re}(z_1 \bar{z}_2 \bar{\omega}^r)$

$= 20 \times 16 + 20 \times 4 + 2 \operatorname{Re}(-z_1 \bar{z}_2)$

$(z_1 \bar{z}_2 = 2|z_2|^2 = 2 \times 4 = 8)$

$= 320 + 80 - 16$

$= 384$

7. **Ans (A,B,C,D)**

$a^2 + b^2 + c^2 + \frac{1}{x^2 y^2 z^2} (x^2 y + xy^2)^2 + (y^2 z + z^2 y)^2 + (xz^2 + z^2 x)^2$

$+ \frac{2}{xyz} \{ayz(y+z) + bxz(x+z) + cyx(x+y)\} = 0$

$\Rightarrow \left(a + \frac{y+z}{x}\right)^2 + \left(b + \frac{x+z}{y}\right)^2 + \left(c + \frac{x+y}{z}\right)^2 = 0$

$\Rightarrow a + \frac{y+z}{x} = 0 \quad b + \frac{x+z}{y} = 0$

$c + \frac{x+y}{z} = 0$

$\Rightarrow ax + y + z = 0$

$x + by + z = 0 \quad x, y, z \neq 0$

$x + y + cz = 0$

This homogenous system of equation has non trivial solution

$\Rightarrow \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$

$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$

$\begin{vmatrix} a-1 & 1-b & 0 \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0$

$(a-1)\{(b-1)c - (1-c)\} - (1-b)\{0 - (1-c)\} = 0$

$(a-1)(b-1) \cdot c - (a-1)(1-c) + (1-b)(1-c) = 0$

$c(1-a)(1-b) + (1-a)(1-c) + (1-b)(1-c) = 0$

$\frac{c}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 0$

$\frac{c}{1-c} + \frac{1}{2} + \frac{1}{1-b} - \frac{1}{2} + \frac{1}{1-a} - \frac{1}{2} = \frac{-1}{2}$

$\frac{1}{2} \left(\frac{1+c}{1-c} \right) + \frac{1}{2} \left(\frac{1+b}{1-b} \right) + \frac{1}{2} \left(\frac{1+a}{1-a} \right) = \frac{-1}{2}$

$\frac{1+c}{1-c} + \frac{1+b}{1-b} + \frac{1+a}{1-a} = -1$

8. Ans (B,C,D)

Let $\alpha_1, \alpha_2, \alpha_3 \dots \alpha_{12}$ be the roots of equation

$$a_{12}x^{12} + a_{11}x^{11} + a_{10}x^{10} + \dots + a_0 = 0$$

$$\alpha_1 + \alpha_2 + \dots + \alpha_{12} = -\frac{a_{11}}{a_{12}} = 4$$

$$\alpha_1 \alpha_2 \dots \alpha_{12} = \frac{1}{3^{12}}$$

$$\frac{\alpha_1 + \alpha_2 + \dots + \alpha_{12}}{12} \geq \sqrt[12]{\alpha_1 \cdot \alpha_2 \dots \alpha_{12}}$$

$$\frac{4}{12} \geq \left(\frac{1}{3^{12}}\right)^{\frac{1}{12}}$$

AM = GM

$$\alpha_1 = \alpha_2 = \alpha_3 \dots \alpha_{12} = \frac{1}{3}$$

$$a_1 = -^{12}C_{11} \times \left(\frac{1}{3}\right)^{11} = -\frac{4}{3^{10}}$$

$$a_2 = ^{12}C_{10} \times \left(\frac{1}{3}\right)^{10} = \frac{22}{3^9}$$

$$a_3 = -^{12}C_9 \times \left(\frac{1}{3}\right)^9 = -\frac{220}{3^9}$$

9. Ans (A,B,C)

$$(a+b)^2 + (a+b+4c)^2$$

$$= (a+b)^2 + (a+2c+b+2c)^2$$

$$\geq (2\sqrt{ab})^2 + (2\sqrt{2ac} + 2\sqrt{2bc})^2$$

$$\geq 4ab + 8ac + 8bc + 16\sqrt{abc^2} \dots (i)$$

$$\Rightarrow ((a+b)^2 + (a+b+4c)^2) \times \left(\frac{a+b+c}{abc}\right) \geq$$

$$\left(\frac{4ab + 8ac + 8bc + 16c\sqrt{ab}}{abc}\right) (a+b+c)$$

$$\geq 8 \left(\frac{1}{2c} + \frac{1}{b} + \frac{1}{a} + \frac{1}{\sqrt{ab}} + \frac{1}{\sqrt{ba}}\right)$$

$$\left(\frac{a}{2} + \frac{a}{2} + \frac{b}{2} + \frac{b}{2} + c\right)$$

$$\geq 8 \times 25 \left(\frac{1}{2a^2b^2c}\right)^{\frac{1}{5}} \left(\frac{a^2b^2c}{2^4}\right)^{\frac{1}{5}}$$

$$\geq 100$$

$$(a+b)^2 + (a+b+4c)^2 \geq 100 \frac{abc}{a+b+c}$$

Maximum value of K is 100

10. Ans (A,D)

$$z_1 = e^{i\alpha} \quad z_2 = e^{i\beta} \quad z_3 = e^{i\gamma}$$

$$|z_1| = 1 \quad |z_2| = 1 \quad |z_3| = 1$$

$$\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} = -1$$

$$z_1^3 + z_2^3 + z_3^3 = -z_1 z_2 z_3$$

$$z_1^3 + z_2^3 + z_3^3 - 3z_1 z_2 z_3 = -4z_1 z_2 z_3$$

$$(z_1 + z_2 + z_3)(z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1) = -4z_1 z_2 z_3$$

$$\text{Let } z_1 + z_2 + z_3 = z$$

$$z^3 - 3z z_1 z_2 z_3 \left(\frac{1}{z_3} + \frac{1}{z_1} + \frac{1}{z_2}\right) = -4z_1 z_2 z_3$$

$$z^3 - 3|z|^2 \cdot z_1 z_2 z_3 = -4 z_1 z_2 z_3$$

$$|z|^3 = |(3|z|^2 - 4)| |z_1| |z_2| |z_3|$$

$$|z|^3 = |(3|z|^2 - 4)|$$

Case-I : $(3|z|^2 - 4) \geq 0$

$$|z|^3 - 3|z|^2 + 4 = 0$$

$$(|z| + 1)(|z| - 2)^2 = 0$$

$$|z| = 2$$

Case-II : $3|z|^2 - 4 < 0$

$$|z|^3 + 3|z|^2 - 4 = 0$$

$$(|z| - 1)(|z| + 2)^2 = 0$$

$$|z| = 1$$

$$|z|^2 \text{ can be 1 or 4}$$

11. Ans (A,C,D)

Here $A = \text{diag}(x + xy + xyz, y + yz + xyz, z + zx + xyz)$

$$A^{-1} = \text{diag} \left(\frac{1}{x + xy + xyz}, \frac{1}{y + yz + xyz}, \frac{1}{z + zx + xyz} \right)$$

and $xyz = 1$

$$\text{Tr}A^{-1} = \frac{1}{x + xy + xyz} + \frac{1}{y + yz + xyz} + \frac{1}{z + zx + xyz}$$

$$= \frac{1}{1 + x + xy} + \frac{1}{y + yz + 1} + \frac{1}{1 + zx + z}$$

$$= \frac{1}{1 + x + xy} + \frac{x}{xy + x + 1} + \frac{1}{1 + xz + z}$$

$$= \frac{1 + x}{1 + x + xy} + \frac{y}{y + 1 + zy}$$

$$= \frac{1 + x}{1 + x + xy} + \frac{xy}{xy + x + xyz} = \frac{1 + x + xy}{1 + x + xy} = 1$$

Sum of maximum and minimum value of $\text{Tr}(A^{-1})$

$$= 1 + 1 = \frac{2}{1}$$

$$p = 2, q = 1$$

12. Ans (A,C,D)

$$\frac{a_1}{1+a_1} = \frac{a_2}{3+a_2} = \frac{a_3}{5+a_3} = \dots$$

$$= \frac{a_{1003}}{2005+a_{1003}}$$

$$\Rightarrow \frac{1+a_1}{a_1} = \frac{3+a_2}{a_2} = \frac{5+a_3}{a_3} = \dots$$

$$= \frac{2005+a_{1003}}{a_{1003}}$$

$$\Rightarrow \frac{1}{a_1} + 1 = \frac{3}{a_2} + 1 = \frac{5}{a_3} + 1 = \dots$$

$$= \frac{2005}{a_{1003}} + 1$$

$$\frac{a_1}{1} = \frac{a_2}{3} = \frac{a_3}{5} = \frac{a_{1003}}{2005} = k$$

$$\Rightarrow a_1 = k, a_2 = 3k, a_3 = 5k, a_4 = 7k \dots a_{1003} = 2005k$$

The sequence of terms are in AP.

Option (C)

$$\text{If } a_4 = 7k = 14 \Rightarrow k = 2$$

$$\text{Then sum of 1st five terms} = a_1 + a_2 + a_3 + a_4 + a_5$$

$$= 2 \times 25 = 50$$

Option (D)

$$\text{If } a_5 = 9k = 27 \Rightarrow k = 3$$

$$\text{Then sum of 1st ten terms} = a_1 + a_2 + a_3 + \dots + a_{10}$$

$$= 3 \times 100 = 300$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (6.00)

$$\text{Let } \omega = e^{\frac{i2\pi}{3}} = e^{-\frac{4\pi}{3}i}$$

$\omega^2 = e^{\frac{i4\pi}{3}} = e^{-\frac{2\pi}{3}i}$, here ω and ω^2 are cube roots of unity.

$$|a + b\omega + c\omega^2|^2 + |a + b\omega^2 + c\omega|^2$$

$$= (a + b\omega + c\omega^2)(\overline{a + b\omega + c\omega^2}) + (a + b\omega^2 + c\omega)(\overline{a + b\omega^2 + c\omega})$$

$$= (a^2 + b^2 + c^2 - ab - bc - ca) + (a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= (a - b)^2 + (b - c)^2 + (c - a)^2$$

$$\geq 6$$

(For minimum value take any three consecutive natural number)

$$\text{Say } a = 1, b = 2, c = 3$$

2. Ans (0.00)

$$(2b - 3c)x + (c - 2a)y + (3a - b)z = 0$$

$$(2c - 3a)x + (a - 2b)y + (3b - c)z = 0$$

$$(2a - 3b)x + (b - 2c)y + (3c - a)z = 0$$

$\Rightarrow$ This can be converted to matrix equation

$$AX = 0$$

$$\text{where } A = \begin{bmatrix} 2b-3c & c-2a & 3a-b \\ 2c-3a & a-2b & 3b-c \\ 2a-3b & b-2c & 3c-a \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2b-3c & c-2a & 3a-b \\ 2c-3a & a-2b & 3b-c \\ 2a-3b & b-2c & 3c-a \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{vmatrix} = 0$$

$\therefore$ Skew symmetric matrix of odd order determinant is zero.

$$\det(A) + \det(2A) + \det(3A) = 0$$

3. Ans (10.00)

$$\text{Apply } R_3 \rightarrow R_3 - R_1$$

$$\Delta = \begin{vmatrix} 1 & 3 \cos \theta & 1 \\ \sin \theta & 1 & 3 \cos \theta \\ 0 & \sin \theta - 3 \cos \theta & 0 \end{vmatrix} = (3 \cos \theta - \sin \theta)^2$$

$$D_{\max} = 10$$

4. Ans (2.50)

$$a^2x^4 - 2abx^3 + (2ac + b^2)x^2 - 2bcx + c^2 = 0$$

$$\Rightarrow (ax^2 - bx + c)^2 = 0$$

$ax^2 - bx + c = 0$ has two distinct roots lies between (0,1)

Let α and β be two distinct real root lies between (0,1)

$$0 < \alpha < 1, 0 < \beta < 1$$

$$\Rightarrow 0 < 1 - \alpha < 1, 0 < 1 - \beta < 1$$

use A.M $\geq$ G.M

$$\left. \begin{aligned} \frac{\alpha + (1-\alpha)}{2} &\geq \sqrt{\alpha(1-\alpha)} \Rightarrow \frac{1}{2} \geq \sqrt{\alpha(1-\alpha)} \\ \frac{\beta + (1-\beta)}{2} &\geq \sqrt{\beta(1-\beta)} \Rightarrow \frac{1}{2} \geq \sqrt{\beta(1-\beta)} \end{aligned} \right\}$$

multiply

$$\Rightarrow \frac{1}{4} > \sqrt{\alpha\beta(1-\alpha)(1-\beta)} \quad (\alpha \neq \beta)$$

$$\alpha\beta(1-\alpha)(1-\beta) < \frac{1}{16}$$

$$\frac{c}{a} \left(1 - \frac{b}{a} + \frac{c}{a} \right) < \frac{1}{16}$$

$$c(a-b+c) < \frac{a^2}{16} \quad \text{where } a-b+c > 0$$

$$\Rightarrow a \geq 5$$

$ax^2 - bx + c = 0$ has real distinct roots

$$b^2 - 4ac > 0$$

$$b^2 > 4ac \Rightarrow b^2 > 20c \Rightarrow c \geq 1$$

$$b \geq 5$$

$$abc \geq 25$$

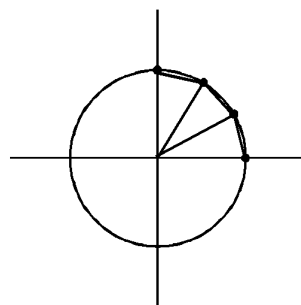
5. Ans (1.10)

$$\begin{aligned} &\sum_{n=2}^{\infty} \frac{n^4 + 3n^2 + 10n + 10}{2^n(n^4 + 4)} \\ &= \sum_{n=2}^{\infty} \left(\frac{n^4 + 4}{2^n(n^4 + 4)} + \frac{3n^2 + 10n + 6}{2^n(n^4 + 4)} \right) \\ &= \sum_{n=2}^{\infty} \frac{1}{2^n} + \sum_{n=2}^{\infty} \frac{3n^2 + 10n + 6}{2^n((n^2)^2 + (2)^2)} \\ &= \frac{1}{4} + \sum_{n=2}^{\infty} \frac{3n^2 + 10n + 6}{2^n(n^2 - 2n + 2)(n^2 + 2n + 2)} \\ &= \frac{1}{2} + \sum_{n=2}^{\infty} \frac{4(n^2 + 2n + 2) - (n^2 - 2n + 2)}{2^n(n^2 - 2n + 2)(n^2 + 2n + 2)} \\ &= \frac{1}{2} + \sum_{n=2}^{\infty} \left(\frac{1}{2^{n-2}(n^2 - 2n + 2)} - \frac{1}{2^n(n^2 + 2n + 2)} \right) \\ &= \frac{1}{2} + \sum_{n=2}^{\infty} \left(\frac{1}{2^{n-2}((n-2)^2 + 2(n-2) + 2)} - \frac{1}{2^n(n^2 + 2n + 2)} \right) \\ &= \frac{1}{2} + \frac{1}{2} + \frac{1}{2 \times 5} \\ &= \frac{1}{2} + \frac{1}{2} + \frac{1}{10} = \frac{11}{10} = 1.10 \end{aligned}$$

6. Ans (3.00)

12th roots of unity excluding 1 are the roots of $x^{11} + x^{10} + \dots + x + 1 = 0$

Now, 1, $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{11}$ are 12th roots of unity.



They lie on an unity circle whose center as origin and they form a regular polygon whose vertices are 1, $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{11}$

$$\begin{aligned} \text{Area} &= 12 \times \frac{1}{2} \times \sin\left(\frac{2\pi}{12}\right) \\ &= 3 \text{ sq. unit.} \end{aligned}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

PART TEST

31-12-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I & I(N)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	D	B	B	C	B	A	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	B	D	B	C	D	B	B	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1.00	19.00	4.65 to 4.67	-12.50	8.80	8.30 to 8.40	3.00	30.00	21.70	2.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	A	B	D	C	B	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	A	B	A	D	A	B	B	C	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	3.00	138.00	40.00	81.00	2.50	2.00	4.00	375.00	0.30	2.92

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	D	B	C	C	C	A	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	A	B	B	B	D	B	C	A	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	3.00	9.00	9.00	2.25	18.00	2.00	1.00	48.00	0.00	-1.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$a_0 \frac{n^2}{Z} = \frac{9}{2} a_0 \Rightarrow n = 3, Z = 2$$

2. Ans (D)

$$E = -c \frac{Z^2}{n^2}$$

3. Ans (D)

$$A = \frac{A_0}{2^n}$$

4. Ans (B)

$$Q = \frac{k \propto 238}{234}$$

5. Ans (B)

$$f \propto \frac{1}{\lambda} = \frac{v}{\lambda}$$

$$= \frac{v}{\frac{1}{Nd^2}} = vNd^2$$

6. Ans (C)

$$Pv^\gamma = a \Rightarrow p \left(\frac{nRT}{P} \right)^\gamma = a$$

$$P^{1-\gamma} (nRT)^\gamma = a$$

$$(10)^5 10^{-5\gamma} \left(10 \times \frac{25}{3} \times 1200 \right)^\gamma$$

7. Ans (B)

$$W_{AB} = 500 \Rightarrow P_A (V_B - V_A)$$

$$W_{CD} = P_C (V_D - V_C)$$

8. **Ans (A)**

$$msd\theta = dQ$$

$$\frac{dm}{dt} s (\Delta\theta) = \frac{dQ}{dt}$$

9. **Ans (B)**

Temperature of Junction increases from T to 2.2T

10. **Ans (D)**

$$P \propto (\text{area}) (T^4 - T_0^4)$$

11. **Ans (D)**

$$T_A = 3T_B$$

$$P = e\sigma 4\pi R^2 T^4$$

12. **Ans (B)**

$$\Delta v = v_o (\gamma - 3 \propto) \Delta\theta$$

13. **Ans (B)**

$$\frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$$

14. **Ans (D)**

$$E = \frac{hC}{\lambda}$$

15. **Ans (B)**

$$\overline{A + \vec{B}} = \vec{A} \cdot \vec{B}$$

16. **Ans (C)**

$$I = \frac{0.7}{3.3} \text{ mA}$$

17. **Ans (D)**

$$\eta = \frac{Q_H - Q_C}{Q_H}$$

18. **Ans (B)**

$$\frac{T_C}{T_H - T_C}$$

19. **Ans (B)**

$$\Delta S = \int_i^f \frac{dQ_r}{T} = nC_V \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}$$

20. **Ans (B)**

$$\text{Heat gain } 1 \times 40 \times \frac{1}{2} = 20$$

$$\text{Heat loss } 1 \times 20 \times 1 = 20$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (1.00)**

System is under thermal Equilibrium only

2. **Ans (19.00)**

$$n \frac{hc}{\lambda} = 2 \times 10^{-3} \times \frac{1}{1.6 \times 10^{-19}}$$

$$n \frac{1240}{620} = 2 \times 10^{-3} \times \frac{10^{-19}}{1.6} \Rightarrow n = \frac{1}{1.6} \times 10^{+16}$$

$$\text{number of e/sec} = \frac{1}{1.6}$$

$$i = 10^{-19} \text{ A}$$

3. **Ans (4.65 to 4.67)**

$$N_x = \frac{N_o}{2^3}$$

$$\Rightarrow N_Q = N_o - \frac{N_o}{2^2}$$

4. **Ans (-12.50)**

$$\frac{5}{R} = \frac{T_4 + 20}{R}$$

5. **Ans (8.80)**

$$\text{Stress} = \gamma \frac{\Delta \ell}{\ell} \Rightarrow \gamma \propto \Delta\theta$$

6. **Ans (8.30 to 8.40)**

$$\frac{10}{5} = -C (55 - 30)$$

$$\frac{10}{t} = -C (45 - 30) \Rightarrow \frac{t}{5} = \frac{25}{15}$$

7. **Ans (3.00)**

$$pr = n \frac{h}{2\pi} \Rightarrow \lambda \propto \frac{r}{n}$$

8. **Ans (30.00)**

$$\frac{c}{\lambda} = a^2 (z - b)^2 \Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{(27 - 1)^2}{(z - 1)^2}$$

$$(z - 1) = \left(\sqrt{\frac{\lambda_1}{\lambda_2}} \right) 26 \Rightarrow (z - 1) = 29 \Rightarrow z = 30$$

9. **Ans (21.70)**

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

10. **Ans (2.00)**

$$\frac{2}{3} = \frac{10 - x}{10 + x} \Rightarrow 30 - 3x = 20 + 2x$$

$$5x = 10 \Rightarrow x = 2$$

PART-2 : CHEMISTRY

SECTION-I

2. **Ans (C)**
Theoretical dipole moment = $1.11 \times 4.8 \text{ D} \%$
Ionic Character
$$= \frac{\text{observed dipole moment}}{\text{theoretical dipole moment}} \times 100$$
3. **Ans (C)**
Factual
4. **Ans (A)**
Covalent character $\propto \phi$
5. **Ans (B)**
 $\text{N}_2^{2-} : \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2 \pi 2p_x^1 \pi^* 2p_y^1$
 $\text{B.O. N}_2^{2-} = \frac{10-6}{2} = 2; \text{B.O. O}_2 = \frac{10-6}{2} = 2$
 NO^- isoelectronic with O_2 so $\text{B.O.} = \frac{10-6}{2} = 2$
All have same number of electrons (i.e. 16) so isoelectronic.
6. **Ans (D)**
 $\text{Bond length} \propto \frac{1}{\text{Bond order}}$
7. **Ans (C)**
Fact
8. **Ans (B)**
$$\frac{Y_A}{Y_B} = \frac{P_A^0}{P_B^0} \times \frac{X_A}{X_B};$$

$$\Rightarrow \frac{X_B}{X_A} = \frac{400}{100} \times \frac{X_A}{X_B} \Rightarrow \frac{X_A}{X_B} = \frac{1}{2}$$

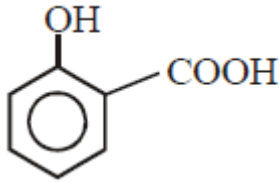
$$\Rightarrow X_A = \frac{1}{3}; X_B = \frac{2}{3}$$
9. **Ans (A)**
Fact (refer module under emulsion)
10. **Ans (B)**
 $2\text{CH}_3\text{COOH} \rightleftharpoons (\text{CH}_3\text{COOH})_2$
$$\begin{matrix} 1 & 0 \\ (1-\alpha) & \frac{\alpha}{2} \end{matrix}$$

 $i = 1-\alpha/2$
or, $\alpha = 2 \times (1-i) = 2(1-0.52) = 0.96$
 $i = 1-0.48 = 0.52$
 $\pi = iCRT$
 $= 0.52 \times 0.1 \times 0.0821 \times 300$
 $= 1.28 \text{ atm}$

11. **Ans (C)**
(A) $\Delta G = \Delta H - T\Delta S < 0$ as $\Delta S < 0$ so ΔH has to be negative
(B) micelles formation will take place above T_k and above CMC
(C) this solution will be negatively charged.
(D) Fe^{3+} ions will have greater flocculability power so smaller flocculating value.
12. **Ans (A)**
B is more volatile than A hence has more tendency to evaporate.
13. **Ans (B)**
Facts about K_H
15. **Ans (D)**
Left ring is more activated than the right one.
16. **Ans (A)**
Arene diazonium cations are reduced by hypophosphorous acid or sodium stannite gives benzene
17. **Ans (B)**
Rate $\propto (+)$ ve density at C attached to halogen
18. **Ans (B)**
Hemi-acetal is present in 'Y', hence reducing sugar.
19. **Ans (C)**
Ozone formed from I, II and IV are identical

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.00)**
I, II, IV are correct
2. **Ans (138.00)**

3. **Ans (40.00)**
 $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$
4. **Ans (81.00)**
 NaSCN is formed.

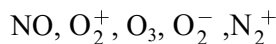
5. **Ans (2.50)**

$$X = 5, Y = 5$$

6. **Ans (2.00)**

2 & 4 are correct

7. **Ans (4.00)**



8. **Ans (375.00)**

For the given solution

$$\Delta T_f = 2 \times K_f \quad \dots(1)$$

560gm solution contains 1 mole urea & 500gm solvent. On placing solution at temperature 5°C below its freezing point then

$$(\Delta T_f + 5) = \frac{1000}{250} \times K_f = 4K_f \quad \dots(2)$$

from (1) & (2)

$$K_f = 5/2$$

$$\Delta T_f = 5$$

$$\text{Now, } \Delta T_f + 15 = K_f m \Rightarrow 2K_f + 15 = K_f m$$

$$\text{or } 8 = m$$

$$\therefore \text{ mass of solvent} = \frac{1}{8} \text{ kg} = \frac{1000}{8} \text{ g} = 125 \text{ g}$$

$$\therefore \text{ mass of solvent} = \text{frozen} = 500 - 125 = 375$$

10. **Ans (2.92)**

$$\frac{P^\circ - P}{P^\circ} = 0.05$$

$$\therefore \frac{P^\circ - P}{P} = \frac{1}{\frac{1}{0.05} - 1} = 0.0526 = \frac{n}{N}$$

$$\text{molality} = \frac{0.0526 \times 1000}{18} = 2.92$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$$|z_1 + z_2 + z_3|^2 = 3^2 + 5 = 14$$

$$\Rightarrow (z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 14$$

$$\Rightarrow |z_1|^2 + |z_2|^2 + |z_3|^2 + (z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1)$$

$$+ (\bar{z}_1z_2 + \bar{z}_2z_3 + \bar{z}_3z_1) = 14$$

$$\Rightarrow 1 + 4 + 9 + 2 \text{Re}(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1) = 14$$

$$\Rightarrow 2 \text{Re}(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1) = 0$$

$$\Rightarrow \text{Re}(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1) = 0$$

2. **Ans (C)**

$$f(x) = I + B, B = \begin{bmatrix} 0 & x & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(f(x))^n = (I + B)^n$$

$$= I^n + {}^nC_1 I^{n-1} B + {}^nC_2 B^2 I^{n-2} \dots + B_n$$

$$= I + {}^nC_1 B \text{ as } B^2 = 0$$

3. **Ans (D)**

$$D = \begin{vmatrix} a & b & 0 \\ b & c & 0 \\ c & a & 0 \end{vmatrix} \begin{vmatrix} a & b & c \\ b & c & a \\ x & y & z \end{vmatrix} = 0$$

4. **Ans (B)**

d = common difference

$$\frac{S_m}{S_n} = \frac{\frac{m}{2} [2a + (m-1)d]}{\frac{n}{2} [2a + (n-1)d]}$$

$$= \frac{m}{n} \frac{(10 + (m-1)d)}{(10 + (n-1)d)}$$

$$= \frac{5((10-d) + md)}{(10-d) + nd}$$

$$\frac{S_m}{S_n} \text{ is independent of } n$$

$$\Rightarrow d = 0, 10$$

$$a_2 = 5, 15$$

5. **Ans (C)**

Let $\cos x = t$

$$\Rightarrow kt^2 - kt + 1 \geq 0 \forall t \in [-1, 1]$$

$$D = k^2 - 4k$$

Case 1:

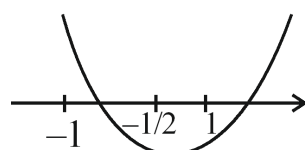
$$k > 0 \text{ and } D \leq 0 \Rightarrow k \in (0, 4] \quad \dots(1)$$

Case 2:

$$k > 0 \text{ and } D > 0$$

$\therefore$ x coordinate of vertex of $y = kx^2 - kx + 1$ is

$$\text{at } x = 1/2$$

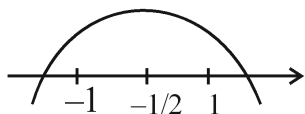


so this case is not accepted

$$\Rightarrow k \in \phi \quad \dots(2)$$

Case 3:

When $k < 0$ & $D > 0 \Rightarrow k \in (-\infty, 0)$



$\Rightarrow f(1) \geq 0 \Rightarrow 1 \geq 0$ which is true & $f(-1) \geq 0$

$\Rightarrow k + k + 1 \geq 0$

$\Rightarrow k \geq -\frac{1}{2}$

so for case 3 : $k \in \left[-\frac{1}{2}, 0\right)$... (3)

So $(1) \cup (2) \cup (3)$

$\Rightarrow k \in \left[-\frac{1}{2}, 0\right) \cup (0, 4]$

also $k = 0$ is special value

which satisfy the inequality

6. **Ans (C)**

$$(A + B)(A - B) = A^2 - B^2$$

$$\Rightarrow AB = BA$$

$$\therefore A^2 B A^{-1} B^{-1} = A \cdot AB \cdot (BA)^{-1}$$

$$= A \cdot BA \cdot (BA)^{-1}$$

$$= A \cdot I$$

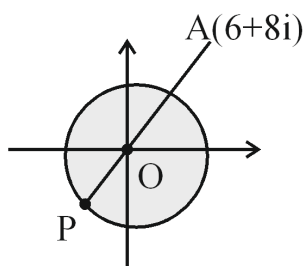
$$= A$$

7. **Ans (C)**

$$|z| \leq 2$$

$$|iz + 6 - 8i| = |i(z - 6i - 8)|$$

$$= |z - (6 + 8i)|$$



$$\text{So, } |z - (6 + 8i)|_{\max} = AP$$

$$= OA + 2$$

$$= 10 + 2 = 12$$

8. **Ans (A)**

$$\begin{vmatrix} a & 5x & p \\ b & 10y & 5 \\ c & 15z & 15 \end{vmatrix} = 125$$

Multiply c_1 with 3

$$\Rightarrow \begin{vmatrix} 3a & 5x & p \\ 3b & 10y & 5 \\ 3c & 15z & 15 \end{vmatrix} = 125 \times 3$$

take 5 common from c_2 & 3 Common from R_3

$$\Rightarrow 5 \times 3 \begin{vmatrix} 3a & x & p \\ 3b & 2y & 5 \\ c & z & 5 \end{vmatrix} = 125 \times 3$$

now interchange rows into columns

$$\Rightarrow \begin{vmatrix} 3a & 3b & c \\ x & 2y & z \\ p & 5 & 5 \end{vmatrix} = 25$$

9. **Ans (C)**

$$S_n = \frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}} = 2 \left(1 - \frac{1}{2^n}\right)$$

$$\therefore 2 - S_n < \frac{1}{100}$$

$$\Rightarrow \frac{1}{2^{n-1}} < \frac{1}{100} \Rightarrow 2^{n-1} > 100$$

$$\Rightarrow n - 1 \geq 7$$

$$\Rightarrow n \geq 8$$

$$n_{\min} = 8$$

10. **Ans (B)**

$$D < 0 \text{ and } n - 2 > 0$$

$$\Rightarrow 64 - 4(n - 2)(n + 4) < 0$$

$$\Rightarrow 16 - (n^2 + 2n - 8) < 0$$

$$\Rightarrow (n + 6)(n - 4) > 0 \text{ and } n > 2$$

the least positive value of $n = 5$

11. **Ans (A)**

$$\text{Let } \frac{a}{|z_2 - z_3|} = \frac{b}{|z_3 - z_1|} + \frac{c}{|z_1 - z_2|} = \lambda \text{ (say)}$$

$$\Rightarrow a = \lambda|z_2 - z_3|, b = \lambda|z_3 - z_1|, c = \lambda|z_1 - z_2|$$

$$a^2 = \lambda^2|z_2 - z_3|^2 = \lambda^2(z_2 - z_3)(\bar{z}_2 - \bar{z}_3)$$

$$\therefore \frac{a^2}{(z_2 - z_3)} = \lambda^2(\bar{z}_2 - \bar{z}_3)$$

Similarly

$$\frac{a^2}{(z_3 - z_1)} = \lambda^2(\bar{z}_3 - \bar{z}_1) \& \frac{c^2}{(z_1 - z_2)} = \lambda^2(\bar{z}_1 - \bar{z}_2)$$

$$\therefore \frac{a^2}{(z_2 - z_3)} + \frac{b^2}{(z_3 - z_1)} + \frac{c^2}{(z_1 - z_2)} = 0$$

12. **Ans (A)**

for unique solution $\Delta \neq 0$

$$\Rightarrow \begin{vmatrix} 2 & a & 6 \\ 1 & 2 & b \\ 1 & 1 & 3 \end{vmatrix} \neq 0$$

$$R_1 \rightarrow R_1 - 2R_3$$

$$R_2 \rightarrow R_2 - R_3$$

$$\Rightarrow \begin{vmatrix} 0 & a-2 & 0 \\ 0 & 1 & b-3 \\ 1 & 1 & 3 \end{vmatrix} \neq 0$$

$$\Rightarrow (a-2)(b-3) \neq 0 \Rightarrow a \neq 2, b \neq 3$$

13. **Ans (B)**

$$T_k = \frac{1}{a + (k-1)d}$$

$$\frac{T_2}{T_6} = 9$$

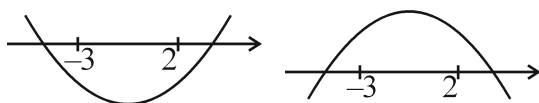
$$\Rightarrow \frac{a+5d}{a+d} = 9 \Rightarrow 8a+4d=0$$

$$\Rightarrow d = -2a$$

$$\frac{T_{10}}{T_4} = \frac{a+3d}{a+9d} = \frac{a-6a}{a-18a}$$

$$= \frac{5}{17}$$

14. **Ans (B)**



$$af(-1) < 0 \text{ and } af(1) < 0, f(-3).f(2) > 0$$

$f(-3)$ Can be negative or positive depending on a

15. **Ans (B)**

$$x^2 - 2x + 4 = 0$$

$$x = 1 \pm i\sqrt{3}$$

$$\alpha = 2(e^{i\frac{\pi}{3}}) \quad \beta = 2e^{-i\frac{\pi}{3}}$$

$$\therefore \alpha^n + \beta^n = 2^n \cdot 2 \cdot \cos\left(\frac{n\pi}{3}\right)$$

16. **Ans (D)**

$$D = 0$$

$$\Rightarrow \begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix} = 0$$

$$\Rightarrow k = \frac{33}{2}$$

17. **Ans (B)**

$$\text{Let } z_1 = \cos \theta + i \sin \theta$$

$$z_2 = 2(\cos \phi + i \sin \phi)$$

$$z_3 = 3(\cos \psi + i \sin \psi)$$

$$\therefore z_1 + z_2 + z_3 = 0$$

$$\Rightarrow z_1^3 + z_2^3 + z_3^3 = 3z_1 z_2 z_3$$

$$\Rightarrow e^{i3\theta} + 8e^{i3\phi} + 27e^{i3\psi} = 18e^{i(\theta+\phi+\psi)}$$

18. **Ans (C)**

$$f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$$

$$f(x) = \frac{x-1}{x+3}; x \neq 2$$

$$\text{range of } f(x) \text{ is } (-\infty, \infty) - \{1, f(2)\}$$

$$(-\infty, \infty) - \left\{1, \frac{1}{5}\right\}$$

19. **Ans (A)**

$$S = 2 + \frac{5}{7} + \frac{8}{49} + \frac{11}{343} + \dots \infty \quad \dots(1)$$

$$S \cdot \frac{1}{7} = \frac{2}{7} + \frac{5}{7^2} + \frac{8}{7^3} + \dots \infty \quad \dots(2)$$

$$(1) - (2)$$

$$S - \frac{S}{7} = 2 + \frac{3}{7} + \frac{3}{7^2} + \frac{3}{7^3} + \dots$$

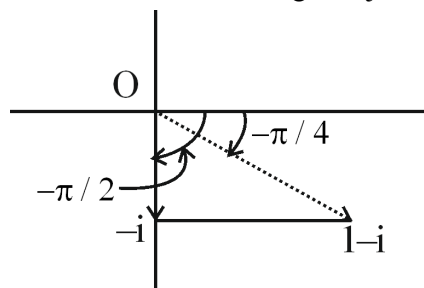
$$\Rightarrow \frac{6S}{7} = 2 + \frac{3}{6} \Rightarrow \frac{6S}{7} = \frac{5}{2}$$

$$\Rightarrow S = \frac{35}{12}$$

20. Ans (C)

$$|Z - 1 + i| + |Z + i| = |(1 - i) - (-i)|$$

⇒ locus of Z is the line segment joining $-i$ with $1 - i$



⇒ minimum arg (Z) = $-\frac{\pi}{2}$ and maximum arg

$$(Z) = -\frac{\pi}{4}$$

⇒ principal arg (Z) $\in \left[-\frac{\pi}{2}, -\frac{\pi}{4}\right]$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (3.00)

$$\begin{bmatrix} 7 & 6 \\ 4 & 7 \end{bmatrix} + b \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} -c & 0 \\ 0 & -c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow b = -2, c = 5$$

2. Ans (9.00)

$$\Delta = 3$$

Required determinant is formed when each elements of first matrix is replace by corresponding co-factors hence $\Delta_{\text{cofactor}} = \Delta^2$

3. Ans (9.00)

$$\alpha^4 - 3\alpha + 1 = 0$$

$$\Rightarrow \alpha^4 = 3\alpha - 1$$

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3$$

$$= \frac{1}{\alpha\beta\gamma\delta} [\alpha^4\beta\gamma\delta + \beta^4\alpha\gamma\delta + \gamma^4\alpha\beta\delta + \delta^4\alpha\beta\gamma]$$

$$= \frac{1}{\alpha\beta\gamma\delta} [(3\alpha - 1)\beta\gamma\delta + (3\beta - 1)\alpha\gamma\delta + (3\gamma - 1)\alpha\beta\delta + (3\delta - 1)\alpha\beta\gamma]$$

$$= \frac{1}{\alpha\beta\gamma\delta} [12\alpha\beta\gamma\delta - \Sigma\alpha\beta\gamma] = \frac{12 - 3}{1} = 9$$

4. Ans (2.25)

$$|z - (1 + i)| = 3|z + 1 + i|$$

$$\sqrt{(x - 1)^2 + (y - 1)^2} = 3\sqrt{(x + 1)^2 + (y + 1)^2}$$

$$\Rightarrow 8x^2 + 8y^2 + 20x + 20y + 16 = 0$$

$$\Rightarrow x^2 + y^2 + \frac{5}{2}x + \frac{5}{2}y + 2 = 0$$

$$Z \text{ lies on circle with radius } r = \sqrt{\frac{25}{16} + \frac{25}{16} - 2}$$

$$r = \frac{3}{2\sqrt{2}}$$

area of square inscribed in circle

$$= 2r^2 = \frac{9}{4}$$

5. Ans (18.00)

$$a + 2b + 3c = 6$$

$$\frac{a + 2b + \frac{3c}{2} + \frac{3c}{2}}{4} \geq \left(a \cdot 2b \cdot \frac{3c}{2} \cdot \frac{3c}{2}\right)^{\frac{1}{4}}$$

$$\Rightarrow \frac{6}{4} \geq \left(\frac{9}{2}abc^2\right)^{\frac{1}{4}}$$

$$\Rightarrow \frac{81}{16} \geq \frac{9}{2}(abc^2)$$

$$\Rightarrow \frac{9}{8} \geq abc^2$$

$$\text{so } abc^2 \text{ max value is } = \frac{9}{8}$$

6. Ans (2.00)

Product of roots < 0

$$\Rightarrow \lambda^2 - \lambda - 2 < 0$$

$$\Rightarrow (\lambda - 2)(\lambda + 1) < 0$$

$$\Rightarrow \lambda \in (-1, 2)$$

so total 2 integer values of λ

7. Ans (1.00)

$$|A^{-1}| = -27 \Rightarrow |A| = -\frac{1}{27}$$

$$|\text{adj}(3A)| = |9\text{adj}A| = 9^3|A|^2$$

$$= 9^3 \frac{1}{27^2} = 1$$

8. **Ans (48.00)**

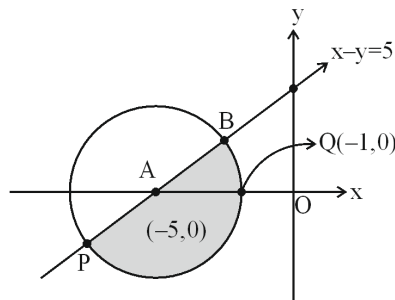
Given $|z + 5| \leq 4$

$$\Rightarrow (x + 5)^2 + y^2 \leq 16 \quad \dots(1)$$

Also, $z(1 + i) + \bar{z}(1 - i) \geq -10$

$$\Rightarrow x - y \geq -5 \quad \dots(2)$$

From (1) and (2) locus of z is the shaded region in the diagram



$|z + 1|$ represents distance of ' z ' from $Q(-1, 0)$

clearly ' P ' is the required position of ' z ' when

$|z+1|$ is maximum

$$\therefore P \equiv (-5 - 2\sqrt{2}, -2\sqrt{2})$$

$$\therefore PQ^2|_{\max} = 32 + 16\sqrt{2}$$

$$\Rightarrow \alpha = 32$$

$$\Rightarrow \beta = 16$$

Thus, $\alpha + \beta = 48$

9. **Ans (0.00)**

Apply $R_3 \rightarrow R_3 - R_1$

$R_2 \rightarrow R_2 - R_1$

$$D = \begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log(r^3) & \log r^3 & \log(r^3) \\ \log(r^6) & \log(r^6) & \log(r^6) \end{vmatrix} = 0$$

where r is common ratio of G.P.

10. **Ans (-1.00)**

$$a_1 = 0, a_2 = 1$$

$$a_n = a_{n-1} - a_{n-2}$$

so sequence is

$$0, 1, 1, 0, -1, -1, 0, 1, 1, 0, \dots$$

so sequence is repeating after every 6 terms

$$2016 = 336 \times 6$$

$$\therefore a_{2016} = a_6 = -1$$